DataMelon Labs

INDUSTRY PROJECT REPORT

DataMelon Labs Industry Fellowship
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INDUSTRY PROJECT - 2026

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This report consolidates the five projects completed during the Industry Project and presents the business problem, analytical work, findings, and business interpretation for each project.

1. Retail Business Performance Deep-Dive - 12-Month Rolling Analysis

Brief Project Summary

An executive-level analysis of retail business performance using a 12-month rolling perspective. The project examined sales growth, loyalty contribution, regional performance, category concentration, payment behavior, customer acquisition, and spend per customer.

Business Problem and Explanation

The objective was to understand the drivers behind overall sales growth rather than relying only on total sales. The analysis focused on identifying where growth was coming from and whether customer acquisition and customer value were moving in the same direction.

Key Work Performed

- Performed 12-month rolling business performance analysis.
- Analyzed sales trends and recent growth performance.
- Evaluated loyalty-program contribution to total sales.
- Analyzed regional contribution and product-category concentration.
- Reviewed payment-method transaction mix.
- Compared customer acquisition growth with spend per customer.
- Built an executive-oriented analytical view of business performance.

Important Insights / Findings

- Total sales reached INR 895.3M.
- Sales showed 7.7% growth over the last 6 months.
- The loyalty program generated INR 497.3M, representing 55% of total sales, and was growing faster than the overall business.
- North contributed 26.5% of sales, while Central contributed 13.9%.
- 16 product categories contributed 80% of sales.
- Card and UPI together represented 93% of transactions.
- Customer acquisition increased by 23.8%, while spend per customer declined by 13% over the last 6 months.

Business Impact / Interpretation

The results indicate that sales growth was supported strongly by customer-base expansion, while individual customer spending declined. This highlights the need for deeper customer-value and cohort analysis to understand how acquisition can be converted into stronger customer value.

2. Like-for-Like (LFL) Performance Analysis

Brief Project Summary

An end-to-end Like-for-Like performance analysis comparing 2026 with 2025 by matching trading days rather than simply comparing calendar dates. The analysis was implemented in Databricks using SQL and organized into Gold-layer tables for Current Year, Previous Year, and LFL comparison.

Business Problem and Explanation

Calendar-date comparisons can be distorted when the number or pattern of trading days differs between periods. The project therefore focused on creating a comparable performance view using matched trading days.

Key Work Performed

- Built the LFL analysis in Databricks using SQL.
- Matched current-year and previous-year trading days.
- Created Gold-layer tables for Current Year, Previous Year, and LFL comparison.
- Analyzed LFL sales and transaction growth.
- Performed regional, city, department, and monthly momentum analysis.

Important Insights / Findings

- Overall LFL sales growth was +19.33%.
- Transaction growth was +20.90%.
- West recorded +21.68% growth, with all regions showing double-digit growth.
- Nagpur grew by +27.13% and Noida by +26.14%.
- Household grew by +32.38%, while core categories remained around 18-20% growth.
- Monthly growth remained in the 18-22% range, with no significant seasonality observed in the analysis.

Business Impact / Interpretation

The LFL approach provides a more comparable view of business performance by controlling for differences in trading days. Growth was broad-based across regions, cities, departments, and months rather than being concentrated in a single area.

3. Geography Deep Dive: Regional Performance Analysis

Brief Project Summary

A geographic performance analysis covering regions, cities, city tiers, store formats, customer acquisition, revenue concentration, and underperforming markets. The project was used to understand geographic growth and the relationship between location, format, and business performance.

Business Problem and Explanation

The business needed to understand whether growth was broad-based geographically, which locations and formats were performing strongly, where underperformance existed, and how revenue concentration could affect geographic strategy.

Key Work Performed

- Analyzed performance across all five regions.
- Reviewed city-level and city-tier performance.
- Compared Fresh, Supermarket, and Express formats.
- Analyzed new-customer growth across geographies.
- Identified revenue concentration among top cities.
- Reviewed product and category demand patterns by geography.
- Identified underperforming cities and formats.

Important Insights / Findings

- There were no declining markets in the analysis; national growth was 20-26% YoY across all five regions.
- West and South showed 25%+ growth.
- Fresh format showed 28-32% growth.
- The top 15 cities contributed 86% of revenue.
- Tier 1 growth was 27.7%, Tier 2 was 22.7%, and Tier 3 was 17.2%.
- The top five cities contributed 39% of revenue.
- Express performance was broadly 15-20% versus 25-30% for Fresh and Supermarket.
- Jabalpur, Raipur, and Dehradun showed 11-17% growth in the Tier 3 Express analysis.
- New-customer growth was 55-65% across the analyzed markets.

Business Impact / Interpretation

The findings support a differentiated geographic strategy: strong growth areas can be studied for expansion, while format-location combinations with weaker performance require closer review. Revenue concentration also makes regional balance and monitoring of major cities important.

4. Department Deep Dive: Product Performance Analysis

Brief Project Summary

A detailed product and department performance analysis covering revenue concentration, category growth and decline, customer segments, regional variation, store formats, seasonality, brands, and customer engagement.

Business Problem and Explanation

The objective was to identify high-performing and underperforming parts of the product portfolio, understand customer and regional differences, and translate the findings into areas for portfolio, merchandising, promotion, and retention analysis.

Key Work Performed

- Analyzed revenue concentration across departments.
- Reviewed category-level growth and decline.
- Segmented customers into Gold, Silver, Bronze and non-loyalty behavioral groups.
- Analyzed regional product variation.
- Compared Hypermarket and Supermarket shopping patterns.
- Reviewed seasonal patterns and peak and lean periods.
- Analyzed brand leaders, challengers, and private-label opportunities.
- Reviewed customer engagement, repeat behavior, and loyalty migration.

Important Insights / Findings

- The top five departments drove 60-70% of total business.
- Growth categories showed 30-50%+ YoY growth in the analysis.
- Repeat customers were identified as 3-5x more valuable than one-time customers.
- Festival periods in October-November showed approximately 30-40% uplift.
- Brand leaders held 40-60% share in the analyzed categories.
- Repeat rate across the analyzed engagement view ranged from 30-90%.
- Hypermarket trips were associated with approximately 12-15 items and INR 1,200-1,500 baskets, while Supermarket trips were associated with 5-8 items and INR 400-600 baskets.

Business Impact / Interpretation

The analysis provides a framework for balancing traffic-driving categories with margin-generating categories, tailoring assortment by region and format, using seasonality in planning, and strengthening customer engagement and retention.

5. Customer Analysis

Brief Project Summary

A customer analytics project focused on customer-base growth, new versus repeat customers, loyalty sales, spend per customer, age groups, regional customer distribution, and monthly customer trends.

Business Problem and Explanation

The business needed to understand whether customer growth was translating into stronger customer value and how customer behavior differed across age groups and regions.

Key Work Performed

- Analyzed current-year customer count and YoY growth.
- Measured new-customer contribution and repeat rate.
- Analyzed loyalty sales and spend per customer.
- Compared age groups using customer count, spend, and transaction behavior.
- Compared customer distribution and spend across regions.
- Reviewed monthly customer growth trends.

Important Insights / Findings

- Current-year customers: 13,944, representing +60.6% YoY growth.
- New customers: 2,453, equal to 17.6% of the total customer base.
- Repeat rate was 96.9%, down 0.4 percentage point versus the previous year.
- Loyalty sales were INR 214.74M, up 48.7% YoY.
- Spend per customer was INR 15,401, down 7.4% YoY.
- Age 36-45 had INR 20,497 spend per customer and 16.23 transactions per customer, with spend per customer up 2.6% YoY.
- Customers aged 60+ had INR 17,767 spend per customer, up 3.3% YoY.
- Age 26-35 was the largest group with 2,120 customers and 30% share; spend per customer was INR 18,621.
- West had 3,580 customers (+64.7%), South 2,826 (+59.8%), and North 2,818 (+61.4%).
- Regional spend per customer was highest in North at INR 15,960, followed by Central INR 15,889, West INR 15,702, East INR 15,377, and South INR 13,472.

Business Impact / Interpretation

The customer base expanded strongly, while spend per customer declined. This indicates that customer-base expansion was an important contributor to sales growth and that customer-value analysis remains important for improving the value generated from the growing base.

6. Overall Project Conclusion

Across the five projects, the analysis provided a connected view of retail business performance, comparable year-over-year growth, geographic performance, product and department performance, and customer demographics.

Overall analytical flow

- **Business Performance:** established the overall sales, loyalty, regional, category, payment, acquisition, and customer-value picture.
- **LFL Growth:** created a comparable year-over-year performance view using matched trading days.
- **Geography:** identified regional, city, tier, and format-level performance differences and revenue concentration.
- **Products and Departments:** evaluated portfolio concentration, category performance, customer segments, formats, seasonality, and brands.
- **Customer Analysis:** examined customer growth, repeat behavior, loyalty sales, spend per customer, age groups, and regions.

Overall Interpretation

The combined analysis shows strong business and customer-base growth across the evaluated views, while also highlighting areas where growth in customer numbers does not necessarily translate into higher spend per customer. The projects provide a structured basis for deeper customer-value, geographic, and product-performance analysis.

End of Report